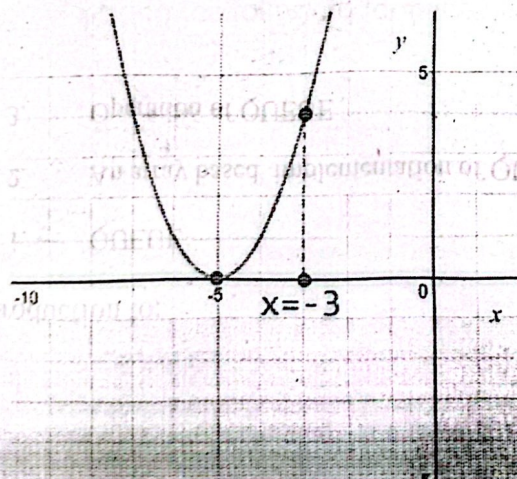


Numerical optimization is the process of finding the best solution to a problem within a given set of constraints. It involves using numerical methods to minimize or maximize a function that represents the objective of the problem.

Numerical methods for optimization include:

1. Gradient Descent: a first-order optimization algorithm that uses the gradient of the function to update the parameters iteratively until convergence.
2. Newton's Method: a second-order optimization algorithm that uses the Hessian matrix of the function to update the parameters iteratively until convergence.
3. Quasi-Newton Methods: a family of optimization algorithms that approximate the Hessian matrix using a sequence of updates based on the gradient of the function.
4. Conjugate Gradient Methods: a family of optimization algorithms that use a set of conjugate directions to update the parameters iteratively until convergence.
5. Simulated Annealing: a stochastic optimization algorithm that uses a random search strategy to find the global minimum of a function.
6. Genetic Algorithms: a stochastic optimization algorithm that simulates the process of natural selection to find the optimal solution.
7. Particle Swarm Optimization: a stochastic optimization algorithm that simulates the behavior of a swarm of particles to find the optimal solution.

These numerical methods are widely used in various fields, such as engineering, finance, machine learning, and data science, to solve complex optimization problems efficiently.



$$y = (x+5)^2$$

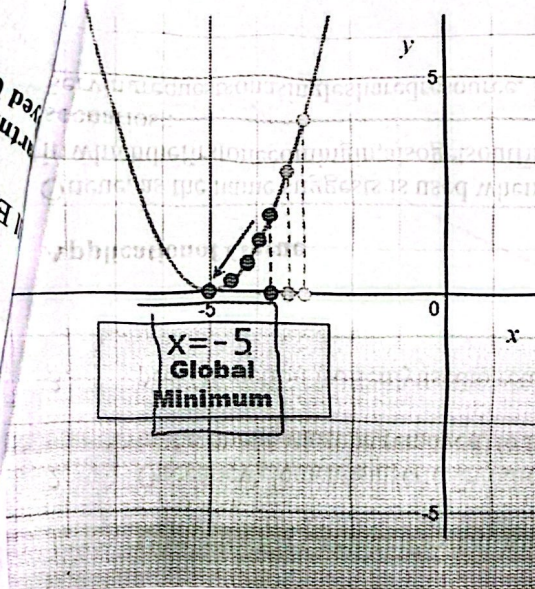
#Step1

Let's start from random point $x = -3$.

Then, find the gradient of the function, $dy/dx = 2 \times (x+5)$

#Step2

Move in the direction of the negative the gradient.



$$y = (x+5)^2$$

#Step3

Perform 2 iterations of gradient descent.

Initialize Parameters

$$X_0 = -3$$

$$dy/dx = 2 \times (x+5)$$

$$\text{learning_rate} = 0.01$$

Iteration 1

$$X_1 = X_0 - (\text{Learning rate}) \times (dy/dx)$$

$$X_1 = (-3) - (0.01) \times (2 \times ((-3)+5)) = -3.04$$

Iteration 2

$$X_2 = X_1 - (\text{Learning rate}) \times (dy/dx)$$

$$X_2 = (-3.04) - (0.01) \times (2 \times ((-3.04)+5)) = -3.0808$$

Example 1:

Let's consider the function $f(x) = x^2$, which we want to minimize using gradient descent.

The first step is to calculate the gradient of the function, which is given by:

$$\nabla f(x) = 2x$$

We start with an initial guess for the minimum value of x , say $x=3$. We then update the value of x using the following formula:

$$x = x - \alpha * \nabla f(x)$$

where α is the learning rate or step size. Let's choose $\alpha = 0.1$ for this example.

The first iteration of gradient descent would be:

$$x = 3 - 0.1 * \nabla f(3) \quad x = 3 - 0.1 * 6 \quad x = 2.4$$

We repeat this process until we reach convergence or the desired accuracy. Let's perform a few more iterations:

$$\text{Iteration 2: } x = 2.4 - 0.1 * \nabla f(2.4) \quad x = 2.4 - 0.1 * 4.8 \quad x = 1.92$$

$$\text{Iteration 3: } x = 1.92 - 0.1 * \nabla f(1.92) \quad x = 1.92 - 0.1 * 3.84 \quad x = 1.536$$

tion 4: $x = 1.536 - 0.1 * \nabla f(1.536) x = 1.536 - 0.1 * 3.072 x = 1.2288$

can see that the value of x is approaching the minimum value of the function, which $= 0$. To check the accuracy of our result, we can calculate the value of the function at final value of x :

$$f(1.2288) = 1.2288^2 = 1.5066$$

This is very close to the minimum value of the function, which is $f(0) = 0$.

Therefore, we can conclude that the minimum value of the function $f(x) = x^2$ is at $x=0$, and we have found this value using gradient descent with a learning rate of 0.1.

Question: Use gradient descent to find the minimum value of the function $f(x) = x^2 + 3x - 5$.

Solution:

Step 1: Calculate the gradient of the function $f(x)$ with respect to x :

$$f'(x) = 2x + 3$$

Step 2: Choose an initial value for x , say $x_0 = 2$.

Step 3: Choose a learning rate, $\alpha = 0.1$.

Step 4: Update x using the gradient descent formula:

$$x_{\{n+1\}} = x_n - \alpha f'(x_n)$$

Plugging in the values:

$$x_{\{1\}} = x_0 - \alpha f'(x_0) = 2 - 0.1(2(2) + 3) = 1.4$$

$$x_{\{2\}} = x_1 - \alpha f'(x_1) = 1.4 - 0.1(2(1.4) + 3) = 0.92$$

$$x_{\{3\}} = x_2 - \alpha f'(x_2) = 0.92 - 0.1(2(0.92) + 3) = 0.568$$

Continuing this process, we can see that x approaches the value of -1.5 as it converges to the minimum of the function.

5: Check if the convergence criteria have been met, such as a maximum number of iterations or a threshold for the change in the function value.

In this example, we could set a threshold for the change in x between iterations or a maximum number of iterations.

Step 6: Verify the solution by checking that the gradient of the function at the minimum point is approximately zero.

$$f'(-1.5) = 2(-1.5) + 3 = 0$$

So, we have found the minimum of the function using the gradient descent algorithm, which is $x = -1.5$.

Hessian matrix

Find the Hessian matrix of the following 2 variable function at the point (1,1):

$$f(x, y) = x^2y + y^2x$$

See solution

First we compute the first-order partial derivatives of the function:

$$\frac{\partial f}{\partial x} = 2xy + y^2$$

$$\frac{\partial f}{\partial y} = x^2 + 2yx$$

then we calculate all the second-order partial derivatives of the function:

$$\frac{\partial^2 f}{\partial x^2} = 2y$$

$$\frac{\partial^2 f}{\partial y^2} = 2x$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = 2x + 2y$$

So the Hessian matrix is defined as follows:

$$H_f(x, y) = \begin{pmatrix} 2y & 2x + 2y \\ 2x + 2y & 2x \end{pmatrix}$$

So the Hessian matrix is defined as follows:

$$H_f(x, y) = \begin{pmatrix} 2y & 2x + 2y \\ 2x + 2y & 2x \end{pmatrix}$$

Finally, we evaluate the Hessian matrix at the point (1,1):

$$H_f(1, 1) = \begin{pmatrix} 2 \cdot 1 & 2 \cdot 1 + 2 \cdot 1 \\ 2 \cdot 1 + 2 \cdot 1 & 2 \cdot 1 \end{pmatrix}$$

$$H_f(1, 1) = \begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix}$$

calculate the Hessian matrix at the point (1,1) of the following function with two variables:

$$f(x, y) = e^{y \ln x}$$

See solution

First we differentiate the function with respect to x and with respect to y:

$$\frac{\partial f}{\partial x} = e^{y \ln x} \cdot \frac{y}{x}$$

$$\frac{\partial f}{\partial y} = e^{y \ln x} \cdot \ln x$$

$$\frac{\partial^2 f}{\partial x^2} = e^{y \ln x} \cdot \frac{y^2}{x^2} - e^{y \ln x} \cdot \frac{y}{x^2}$$

$$\frac{\partial^2 f}{\partial y^2} = e^{y \ln x} \cdot \ln^2 x$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = e^{y \ln x} \cdot \frac{y}{x} \cdot \ln x + e^{y \ln x} \cdot \frac{1}{x}$$

So the Hessian matrix of the function is a square matrix of order 2:

$$H_f(x, y) = \begin{pmatrix} e^{y \ln x} \cdot \frac{y^2}{x^2} - e^{y \ln x} \cdot \frac{y}{x^2} & e^{y \ln x} \cdot \frac{y}{x} \cdot \ln x + e^{y \ln x} \cdot \frac{1}{x} \\ e^{y \ln x} \cdot \frac{y}{x} \cdot \ln x + e^{y \ln x} \cdot \frac{1}{x} & e^{y \ln x} \cdot \ln^2 x \end{pmatrix}$$

And we evaluate the Hessian matrix at the point (1,1):

And we evaluate the Hessian matrix at the point (1,1):

$$H_f(1,1) = \begin{pmatrix} e^{1 \ln(1)} \cdot \frac{1^2}{1^2} - e^{1 \ln(1)} \cdot \frac{1}{1^2} & e^{1 \ln(1)} \cdot \frac{1}{1} \cdot \ln(1) + e^{1 \ln(1)} \cdot \frac{1}{1} \\ e^{1 \ln(1)} \cdot \frac{1}{1} \cdot \ln(1) + e^{1 \ln(1)} \cdot \frac{1}{1} & e^{1 \ln(1)} \cdot \ln^2(1) \end{pmatrix}$$

$$H_f(1,1) = \begin{pmatrix} e^0 \cdot 1 - e^0 \cdot 1 & e^0 \cdot 1 \cdot 0 + e^0 \cdot 1 \\ e^0 \cdot 1 \cdot 0 + e^0 \cdot 1 & e^0 \cdot 0 \end{pmatrix}$$

$$H_f(1,1) = \begin{pmatrix} 1 - 1 & 0 + 1 \\ 0 + 1 & 1 \cdot 0 \end{pmatrix}$$

$$H_f(1,1) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$